

Madhavaram Krishna Gopal Rao

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Summary

Software Engineer with 4+ years of experience building enterprise applications using **Java**, **Spring Boot**, **JavaScript**, **Python**, and **SQL**. Strong background in full-stack development, microservices, distributed systems, workflow automation, and cloud-based production support. Experienced in building claims, payroll, and enterprise data platforms with a focus on performance, secure integrations, observability, and reliable delivery in high-volume business environments.

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL

Backend: Spring Boot, Spring MVC, Spring Security, Spring Batch, Hibernate/JPA, REST APIs, SOAP Services

Frontend: React, Angular, HTML, CSS

AI & Observability: OpenAI API, Rule-Based Processing, OpenTelemetry, AWS CloudWatch

Architecture: Microservices, Distributed Systems, Event-Driven Architecture, Workflow Automation, System Integration

Databases: Oracle, PostgreSQL, MySQL, MongoDB, Redis, AWS RDS

Messaging & Data: Confluent Kafka, ETL Pipelines, Batch Processing, Data Validation, Data Cleansing

Cloud & DevOps: AWS, Docker, Kubernetes, OpenShift, Jenkins, GitHub Actions, CI/CD

Security & Testing: OAuth2, JWT, JUnit, Mockito, Integration Testing, SSO, Role-Based Access Control

Professional Experience

State Farm Insurance

Bloomington, Illinois

Software Engineer

June 2024 – Present

- Built **Spring Boot** services for claims workflows, reducing manual processing effort by **65%** through automated validation, routing, and business-rule checks.
- Integrated the **OpenAI API** into claims support workflows to generate structured summaries from intake details, improving response accuracy and helping downstream teams process claims faster.
- Developed **Kafka**-based services for claim intake and request routing, improving response latency by **40%** for high-volume claims workflows.
- Developed **REST APIs** for internal claims applications and supported **AWS**-based deployment workflows using **RDS** for relational data access and **CloudWatch** for production monitoring.
- Improved observability by integrating **OpenTelemetry** and centralized logging, reducing issue detection and troubleshooting time in production.
- Optimized **SQL** queries and indexing for large claims datasets by improving filter conditions, join performance, and indexed lookup paths, reducing query execution time by **60%** and Debugged production issues.

ADP

Hyderabad, India

Software Engineer

June 2022 – Aug 2023

- Developed **Spring Boot** microservices for payroll and business-system workflows, improving scalability and maintainability of backend processing.
- Built secure **REST APIs** for client data ingestion using **OAuth2** and **JWT**, improving reliability of downstream service integrations.
- Implemented **Confluent Kafka** for asynchronous messaging between distributed services, supporting high-volume payroll processing and reducing service coupling.
- Automated recurring payroll data operations using **Spring Schedulers** and batch processing, improving throughput for scheduled workflows.
- Used **MongoDB** for document-based storage and **Redis** for caching frequently accessed data, improving response times for read-heavy operations.
- Applied the **Token Bucket** algorithm for API rate limiting and worked with gateway-based request handling to improve stability during peak traffic.
- Supported CI/CD workflows using **Jenkins** and **OpenShift** to improve release consistency and deployment reliability.

Cognizant

Hyderabad, India

Software Engineer

April 2021 – May 2022

- Worked on development of **Java** and **Spring Boot** backend services for enterprise transaction and business processing systems.
- Worked on development of **ETL** and data processing pipelines to extract, transform, and load large datasets across enterprise systems, improving data availability and consistency.
- Automated data validation and cleansing workflows using staging checks, field-level validation rules, and transformation logic, reducing manual effort and improving downstream data quality.
- Optimized **Oracle** queries, stored procedures, and indexing strategies for ETL-heavy workloads, reducing query latency by **50%** and improving batch processing performance.
- Developed **REST** and **SOAP** integrations to connect modern services with legacy enterprise platforms and support cross-system communication.

Iowa State University

Ames, Iowa

Graduate Research Assistant

Aug 2023 – Jun 2024

- Built real-time data processing pipelines in **Python** for motion tracking and pose estimation workflows.
- Developed algorithmic workflows for motion analysis using computer vision and numerical validation, improving pose accuracy by **95%** and reducing system latency by **60%**.

Education

Iowa State University

Ames, IA

Master of Science in Computer Science

Aug 2023 – Aug 2025

Vardhaman College of Engineering

Hyderabad, India

Bachelor of Technology in Computer Science

May 2018 – June 2022